Name:V.Naga Revanth

Reg no:192411128

Course code:CSA1007

Course name: Software Engineering

Project: Intelligent Nuclear Power Plant Monitoring Platform

Project Vision, Objectives, and Stakeholders

1. Project Vision

The Intelligent Nuclear Power Plant Monitoring Platform is designed to improve the safety, reliability, and efficiency of nuclear power plant operations. The system uses IoT sensors, Artificial Intelligence (AI), Machine Learning (ML), and real-time monitoring technologies to continuously observe reactor conditions, detect operational anomalies, predict equipment failures, monitor radiation levels, and support emergency response activities.

The platform aims to provide plant operators and management with accurate, real-time information for better decision-making while ensuring compliance with nuclear safety regulations.

Project Objectives

1. Monitor reactor conditions continuously.
2. Detect operational anomalies in real time.
3. Predict equipment failures before breakdowns occur.
4. Track radiation levels throughout the plant.
5. Support emergency response activities.
6. Generate compliance and safety analytics reports.

Project Scope

The system includes real-time monitoring, predictive maintenance, radiation monitoring, emergency alerts, data analytics, and report generation. It focuses on decision support and safety enhancement rather than direct reactor control.

2. Stakeholder Identification

Stakeholder	Responsibilities
Plant Operators	Monitor reactor conditions and respond to alerts.
Control Room Engineers	Supervise plant operations and analyze system data.
Maintenance Team	Perform preventive and corrective maintenance.
Safety Officers	Monitor radiation exposure and ensure safety compliance.
Plant Management	Review reports and make strategic decisions.
Regulatory Authorities	Verify compliance with safety regulations.
Emergency Response Team	Handle emergencies and incident response.
Software Development Team	Design, develop, test, and maintain the platform.
Cybersecurity Team	Protect system data and infrastructure.

Key Challenges

- Managing large volumes of sensor data.
- Detecting failures before they occur.
- Preventing human errors during monitoring.
- Ensuring radiation safety.
- Maintaining system reliability and security.
- Meeting strict regulatory requirements.

Epics, User Stories, and Acceptance Criteria

3. Epics and User Stories

Epic 1: Reactor Monitoring

User Story 1
As a Plant Operator, I want to monitor reactor temperature, pressure, and coolant levels in real time so that I can identify unsafe conditions quickly.

User Story 2
As a Control Room Engineer, I want a live monitoring dashboard so that I can make informed operational decisions.

Epic 2: Anomaly Detection

User Story 3
As a Safety Officer, I want the system to automatically detect abnormal reactor conditions so that safety risks can be minimized.

User Story 4
As a Plant Operator, I want instant alerts whenever abnormal readings occur so that immediate action can be taken.

Epic 3: Predictive Maintenance

User Story 5
As a Maintenance Engineer, I want AI-based failure prediction so that preventive maintenance can be scheduled before equipment failure occurs.

Epic 4: Radiation Monitoring

User Story 6
As a Safety Officer, I want continuous radiation monitoring so that workers remain safe.

Epic 5: Compliance Reporting

User Story 7
As a Regulatory Officer, I want automatic compliance reports so that inspections can be performed efficiently.

4. Acceptance Criteria

User Story 1

Given reactor sensors are active
When the operator opens the dashboard
Then real-time temperature, pressure, and coolant data must be displayed.

User Story 3

Given abnormal reactor conditions occur
When the AI engine detects the anomaly
Then an alert must be generated within 5 seconds.

User Story 5

Given equipment historical data is available
When predictive analysis is performed
Then maintenance recommendations must be generated.

User Story 6

Given radiation exceeds the safety threshold
When the system identifies the condition
Then emergency notifications must be sent immediately.

User Story 7

Given monitoring data is available
When a report is requested
Then a compliance report must be generated in PDF format.

Product Backlog and Sprint Backlog

5. Product Backlog Prioritization

Priority	Backlog Item	Business Value
1	User Authentication	High
2	Sensor Data Collection	High
3	Reactor Monitoring Dashboard	High
4	Real-Time Alert System	High
5	AI Anomaly Detection	High
6	Radiation Monitoring	High
7	Predictive Maintenance	Medium
8	Compliance Report Generation	Medium
9	Historical Data Analysis	Medium
10	Advanced Analytics Dashboard	Low

Prioritization Rationale

High-priority items are selected first because they provide the core functionality needed for monitoring reactor conditions and ensuring safety. Advanced analytical features are scheduled for later sprints.

6. Sprint Backlog (Sprint 1)

Sprint Duration

2 Weeks

Selected User Stories

1. User Authentication
2. Sensor Data Collection

3. Reactor Monitoring Dashboard
4. Real-Time Alert System

Task Breakdown

User Authentication

- Design login page.
- Create user database.
- Implement role-based access control.
- Test authentication features.

Sensor Data Collection

- Integrate reactor sensors.
- Collect temperature data.
- Collect pressure data.
- Store sensor readings in the database.

Monitoring Dashboard

- Design dashboard interface.
- Display live reactor parameters.
- Create charts and graphs.
- Test dashboard updates.

Alert System

- Define threshold values.
- Implement notification mechanism.
- Generate warning messages.
- Test alert functionality.

Story Point Estimation, Sprint Goal, and Deliverables

7. Story Point Estimation

Estimation Technique: Fibonacci Sequence

User Story	Story Points	Justification
User Authentication	3	Simple functionality with standard implementation.
Sensor Data Collection	8	Requires IoT integration and real-time data handling.
Monitoring Dashboard	5	Moderate UI and visualization effort.
Real-Time Alerts	5	Requires event detection and notifications.
AI Anomaly Detection	13	Complex AI model development and testing.
Predictive Maintenance	13	Advanced machine learning implementation.
Radiation Monitoring	8	Sensor integration and safety validation.
Compliance Reports	5	Report generation and export features.

Justification

The Fibonacci estimation method is used because it effectively represents increasing complexity and uncertainty. AI-based functionalities receive higher story points due to their technical complexity and testing requirements.

8. Sprint Goal

The primary goal of Sprint 1 is to develop the foundation of the Intelligent Nuclear Power Plant Monitoring Platform by implementing secure user authentication, real-time sensor data collection, monitoring dashboards, and alert functionality.

9. Expected Sprint Deliverables

At the end of Sprint 1, the following deliverables will be completed:

- User Authentication Module
- Secure Login System
- Real-Time Sensor Data Collection Module
- Reactor Monitoring Dashboard
- Live Temperature and Pressure Monitoring
- Real-Time Alert and Notification System

- Database Integration
- Initial Testing Report
- Sprint Review and Demonstration Document

Conclusion

The Agile Sprint Planning process provides a structured approach for developing the Intelligent Nuclear Power Plant Monitoring Platform. Through well-defined epics, user stories, acceptance criteria, backlog prioritization, sprint planning, and effort estimation, the project team can deliver a reliable, scalable, and secure monitoring solution. The proposed Sprint 1 establishes the foundation for future enhancements such as AI-based anomaly detection, predictive maintenance, radiation analytics, and compliance reporting, ultimately improving nuclear plant safety and operational efficiency.